Megaloblastic — Treatment

1. B12 FIRST warning

WHAT YOU SEE

CENTER 'RED WARNING B12 FIRST', brain

WHAT IT ENCODES

THE single most tested treatment rule: in any megaloblastic anemia, replete/exclude B12 BEFORE giving folate. Folic acid alone in an undiagnosed B12 deficiency floods the thymidine pathway and corrects the anemia, but does nothing for the methylmalonyl-CoA/myelin pathway — so it can PRECIPITATE or accelerate irreversible subacute combined degeneration. When in doubt, give BOTH (B12 + folate) until B12 status is confirmed. Mnemonic: 'B12 before folate, or you fix the blood and wreck the cord.'

BOARD TRAP

Starting folate first because it 'corrects the anemia faster.' This is the classic wrong answer: folate masks the hematologic deficiency while permitting B12 neuro damage (subacute combined degeneration) to progress, sometimes irreversibly. Correct move: check/treat B12 first, or give B12 and folate together.

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2. Folate fixes anemia, hides neuro

WHAT YOU SEE

'FIXES ANEMIA HIDES NEURO'

WHAT IT ENCODES

Folate supplies enough one-carbon units to bypass the methylfolate trap for DNA synthesis, so the megaloblastic anemia and macrocytosis improve — but the B12-dependent methylmalonyl-CoA mutase reaction is untouched, so myelin damage continues silently ('hides' the neuro disease behind a normal CBC). This is why neuro symptoms can WORSEN even as the hemoglobin rises. Always document the neuro exam and B12/MMA before folate. Mnemonic: 'folate paints over the warning light — the engine still burns.'

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3. IM hydroxocobalamin loading

WHAT YOU SEE

LEFT HANGAR 'HYDROXO' drum, 'q3 MO', loading doses

WHAT IT ENCODES

Hydroxocobalamin IM is the preferred parenteral form: typical loading is 1000 µg IM on alternate days for ~1-2 weeks (or longer if neuro signs are present) until improvement, then maintenance 1000 µg IM every ~2-3 MONTHS. It is preferred over cyanocobalamin because it is retained longer (higher protein binding) → fewer injections. (Hydroxocobalamin is also the antidote for cyanide poisoning.) Mnemonic: 'Hydroxo = Held longer → q3-monthly maintenance.'

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4. Cyanocobalamin alternative

WHAT YOU SEE

'CYANO' drum 'q1 MO'

WHAT IT ENCODES

Cyanocobalamin is an acceptable alternative but is cleared faster, so maintenance is roughly MONTHLY (1000 µg IM q1 month) rather than every 3 months. Same loading concept (frequent doses until repletion). Either form fully corrects deficiency; choice is mostly availability/retention. Mnemonic: 'Cyano = Cleared quick → q1-monthly.'

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5. Lifelong if pernicious/malabsorption

WHAT YOU SEE

'LIFELONG MISSION', 'LARGER RETENTION'

WHAT IT ENCODES

When the cause is IRREVERSIBLE malabsorption — pernicious anemia, total/partial gastrectomy or gastric bypass, ileal resection/Crohn — B12 replacement is LIFELONG, because the absorption defect never resolves. By contrast, dietary deficiency (e.g., strict vegan) can be maintained with oral supplements once stores are restored. Stopping therapy in PA reliably causes relapse, including neuro relapse. Mnemonic: 'broken pump = lifelong refills.'

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6. High-dose oral B12 option

WHAT YOU SEE

'CORRECTABLE', oral high-dose, tapeworm/sprue surgery

WHAT IT ENCODES

High-dose ORAL B12 (1000-2000 µg/day) is effective even in PA, because ~1% of an oral dose is absorbed passively by mass action, independent of intrinsic factor — trials show it matches IM for repletion in stable patients. IM is still preferred for severe deficiency, neuro involvement, malabsorption you can't ensure compliance with, or initial loading. Reversible causes (fish tapeworm — treat with praziquantel; tropical sprue/celiac; bacterial overgrowth) can have therapy stopped once cured. Mnemonic: 'oral works in PA — 1% sneaks in without IF.'

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7. Folic acid lifelong/chronic

WHAT YOU SEE

FOLATE bottle 'LIFELONG', MTX/TMP block

WHAT IT ENCODES

Folate deficiency is repleted with oral folic acid ~1-5 mg/day; deficiency corrects within weeks and may be continued chronically when the cause persists (chronic hemolysis e.g. sickle cell, pregnancy, malabsorption, chronic antifolate drugs). For deficiency caused by dihydrofolate-reductase inhibitors (methotrexate, trimethoprim, pyrimethamine), the rescue agent is FOLINIC acid (leucovorin, already-reduced folate), NOT plain folic acid, because the blocked enzyme cannot reduce folic acid. Mnemonic: 'DHFR blocked → give the already-reduced one (folinic/leucovorin).'

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8. Watch potassium (refeeding)

WHAT YOU SEE

TRAP 1 'K+ MINE', 'new cells suck up potassium', 'monitor K+'

WHAT IT ENCODES

As repletion triggers an explosive burst of erythropoiesis, the new cells rapidly take up potassium, causing HYPOKALEMIA (occasionally severe enough to precipitate arrhythmias/sudden death) within the first days — a recognized complication of treating severe megaloblastic anemia. Monitor and replace K+ during early repletion. Brisk cell turnover can also unmask iron deficiency and (less commonly) drop phosphate. Mnemonic: 'new red cells are hungry — they eat potassium (and iron).'

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9. Reticulocytosis 1-2 weeks

WHAT YOU SEE

TRAP 2 'PLT EXPLOSION', 'reticulocyte peak 1-2 wk'

WHAT IT ENCODES

Expected response confirms the diagnosis: reticulocytosis begins by day 2-3 and PEAKS at ~5-7 days (up to ~1-2 weeks), with hemoglobin normalizing over ~6-8 weeks and hypersegmented neutrophils clearing over ~2 weeks. A brisk reticulocyte response after empiric B12/folate is both therapeutic and diagnostic. Failure to respond should prompt rechecking the diagnosis (e.g., concomitant iron deficiency, MDS, hypokalemia limiting response). Mnemonic: 'retic peak at one week = right diagnosis.'

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10. Avoid over-transfusion

WHAT YOU SEE

TRAP 3 'TRANSFUSION QUICKSAND', 'slow 1-2 units'

WHAT IT ENCODES

In chronic severe megaloblastic anemia the patient is euvolemic/expanded (plasma volume compensates), so aggressive transfusion risks transfusion-associated circulatory overload (TACO) and pulmonary edema. If transfusion is truly needed for cardiac compromise, give SLOWLY, 1 unit at a time, often with a loop diuretic, and reassess. Vitamin repletion — not transfusion — is the definitive treatment. Mnemonic: 'go slow — the tank is already full of plasma.'

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11. Severe / emergency patient

WHAT YOU SEE

RIGHT 'SEVERELY ILL PATIENT', emergency sequence

WHAT IT ENCODES

For severe, symptomatic deficiency (heart failure, profound anemia, neuro deficits): draw B12, folate, MMA and homocysteine FIRST, then start parenteral B12 1000 μg IM (plus folate) empirically without waiting for results, replace potassium prophylactically, and transfuse only cautiously if cardiac compromise. Treat neuro cases longer/more intensively as deficits can be partially irreversible. Mnemonic: 'draw, then dose — don't let the patient deteriorate awaiting labs.'

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12. Iron co-supplementation

WHAT YOU SEE

'+ HF Rx', iron

WHAT IT ENCODES

The surge in erythropoiesis after B12/folate repletion rapidly consumes iron stores and can UNMASK or precipitate iron deficiency, blunting the hemoglobin response. Monitor iron studies during recovery and supplement iron if ferritin/transferrin saturation falls — a dimorphic anemia (combined macro- + microcytic) may have a normal-looking MCV that splits as treatment proceeds. Mnemonic: 'feed the new factory iron, or the response stalls.'

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13. Don't wait for results

WHAT YOU SEE

STEP 3 'DO NOT WAIT FOR RESULTS'

WHAT IT ENCODES

In an emergency, after drawing baseline levels (B12, folate, MMA, homocysteine), start empiric B12 + folate immediately — delaying treatment for lab turnaround risks irreversible neurologic injury and cardiopulmonary decompensation. Giving both vitamins together is safe and avoids the folate-first trap. Mnemonic: 'sample first, then treat now — never delay B12 for a pending level.'
